assignment circular motion - more vectors with calculus - bonus ahead

This is a preview of the published version of the quiz

Started: Apr 29 at 7:31pm

Quiz Instructions

2 attempts. There is a tutorial in dropbox.

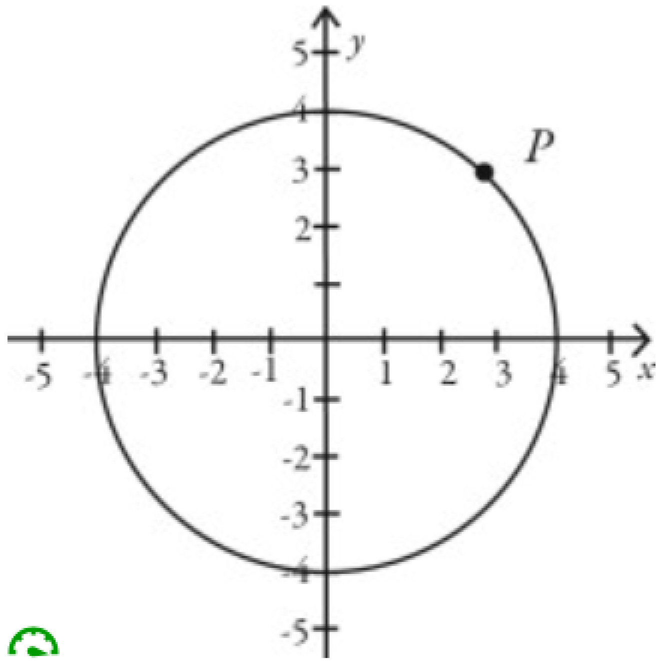
Question 1 4 pts

When an object moves in uniform circular motion, the direction of its acceleration is

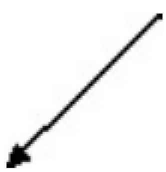
- is directed toward the center of its circular path.
- depends on the speed of the object.
- is directed away from the center of its circular path.
- in the same direction as its velocity vector.
- in the opposite direction of its velocity vector.

Question 2 4 pts

Point *P* in the figure indicates the position of an object traveling at constant speed clockwise around the circle. Which arrow best represents the direction of the acceleration of the object at point *P*?



y
x
P
5
4
3
2
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 1 2 3 4 5



Question 3 4 pts

An aircraft performs a maneuver called an "aileron roll." During this maneuver, the plane turns like a screw as it maintains a straight flight path, which sets the wings in circular motion. If it takes it 35 s to complete the circle and the wingspan of the plane is 11 m, what is the acceleration of the wing tip?

0.99 m/s^2

0.18 m/s^2

1.0 m/s^2

5.6 m/s^2

Question 4 4 pts

A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

67.9 m/s^2

38.7 m/s^2

29.3 m/s^2

14.8 m/s^2

74.2 m/s^2

Question 5 6.3 pts

An electrical motor spins at a constant 2857.0 rev/min. If the armature radius is 2.685 cm what is the acceleration of the outer edge of the armature?

2403 m/s^2

○ 844.4 m/s^2

○ $241{,}100 \text{ m/s}^2$

○ 84.40 m/s^2

Question 6 6.3 pts

You are making a circular turn in your car on a horizontal road when you hit a big patch of ice, causing the force of friction between the tires and the road to become zero. While the car is on the ice, it

○ moves along a straight-line path away from the center of the circle.

○ moves along a straight-line path in its original direction.

○ continues to follow a circular path, but with a radius larger than the original radius.

○ moves along a path that is neither straight nor circular.

○ moves along a straight-line path toward the center of the circle.

Question 7 6.3 pts

When a car goes around a circular curve on a horizontal road at constant speed, what force causes it to follow the circular path?

○ the friction force from the road

○ gravity

○ the normal force from the road

○ No force causes the car to do this because the car is traveling at constant speed and therefore has no acceleration.

Question 8 4 pts

For an object in uniform circular motion, its velocity and acceleration vectors are always perpendicular to each other at every point in the path. (slide 69)

○ True

○ False

Question 9 6.3 pts

If an object travels at a constant speed in a circular path, the acceleration of the object is
(ac is proportional to speed squared and inversly proportional to the radius)

- in the opposite direction of the velocity of the object.
- smaller in magnitude the smaller the radius of the circle.
- in the same direction as the velocity of the object.
- zero.
- larger in magnitude the smaller the radius of the circle.

Question 10 6.3 pts

A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

- 38.7 m/s^2
- 29.3 m/s^2
- 14.8 m/s^2
- 67.9 m/s^2
- 74.2 m/s^2

Question 11 6.3 pts

You are making a circular turn in your car on a horizontal road when you hit a big patch of ice, causing the force of friction between the tires and the road to become zero. While the car is on the ice, it

- moves along a path that is neither straight nor circular.
- moves along a straight-line path toward the center of the circle.
- continues to follow a circular path, but with a radius larger than the original radius.

○ moves along a straight-line path away from the center of the circle.

○ moves along a straight-line path in its original direction.

Question 12 6.3 pts

A car is moving along a circle of radius R and its speed is V. If the car increases its speed by a factor of 3 (x3) , the centripetal acceleration (or force) is (suppose R stays the same)
hint: you can suppose V=1m/s , R = 1m. Find a1. Then take V = 3m/s, R = 1m find a2.
What is a2/a1

○ x1

○ x3

○ stays the same

○ x 9

Question 13 6.3 pts

A car is moving along a circle of radius R and its speed is V. If the car decreases its speed by a factor of 2 (divided by 2) , the centripetal acceleration (or force) is (suppose R stays the same)
hint: you can suppose V=4m/s , R = 1m. Find a1. Then take V = 2m/s, R = 1m find a2

○ divided by 6

○ divided by2

○ stays the same

○ divided by 4

Question 14 6.3 pts

A car is moving along a circle of radius R and its speed is V. If for the same speed, the radius is divided by 3 then the acceleration
hint: you can suppose V = 1m/s, R =6m find a1. Then.. you get it.

○ is multiplied by 3

○ is multiplied by 9

○ is divided by 3

divided by 6

Question 15 6.3 pts

A satellite orbits the earth a distance of 1.50×10^7 m above the planet's surface and takes 8.65 hours for each revolution about the earth. The earth's radius is 6.38×10^6 m. The acceleration of this satellite is closest to
hint: you need to add the altitude to the EArth radius to find r (distance Earth center to satellite). Use slide 91 equation. 8.65 hours is the period T. convert to seconds.

- 0.870 m/s^2.
- 9.80 m/s^2.
- 2.72 m/s^2.
- 1.91 m/s^2.
- 0.0690 m/s^2.

Question 16 6.3 pts

An object moves in a circle of radius *R* at constant speed with a period *T*. If you want to change only the period in order to cut the object's acceleration in half, the new period should be
hint: use equation from slide 90. r is constant. Say 1m. Say T =2s what is a1? Then take a2=a1/2. What happens to T for the same r. T2 = T .... (or use Algebra).

- $T\sqrt{2}$.
- *T*/4.
- $T/\sqrt{2}$.
- *T*/2.
- 4*T*.

Question 17 4 pts

One rope pulls a barge directly east with a force of 41 newtons, and another rope pulls the barge directly north with a force of 89 newtons

What is the vector sum? Use polar coordinate.

hints: -> draw the vectors in a (X,Y) system. Draw the sum using tail to head (or parallelogram).

You see its easy.

-> find magnitude and direction.

- 130 N @ 25 degrees E of N
- 48 N @ 55 E of N
- 98 N @ 65 degrees N of E
- 98 N @ 65 N of E

Question 18 4 pts

) A hot-air balloon is rising vertically 11 ft/sec while the wind is blowing horizontally at 6 ft/sec. Find the angle that the balloon makes with the horizontal. Round to the nearest tenth of a degree.

hint: velocity is a vector

-> draw a balloon at the origin of (X,Y). Use tail to head: draw the velocity of the ballon first (up) then on its head draw the velocity of the wind (right).

> connect the origin to the head of the velocity of the wind. That's your sum. That will be the velocity of the balloon as seen from the ground.The tail is at the origin.

-> find the direction of this vector sum relative to the + x-axis (CCW)

- 61.4 degrees
- 55.3 degrees

- 49 degrees
- 28.6 degrees

Question 19 4 pts

A plane is heading due south with an airspeed of 268 mph. A wind from a direction of 52.0° is blowing at 13.0 mph.

The velocity of the plane / ground has an angle of ... relative to the +xaxis (relative to east)

its 52 N of E or just + 52 degrees.

hint:

-> draw the 2 vectors in a (X,Y) system and attach the tails to the origin. Daw the sum of the 2 vectors (parallelogram method). The tail of the sum is attached to the (0,0).

-> You can easily find the components of the sum looking at your picture. If you don't see it then just find the components of the individual vectors and add them. Sumx = V1x+V2x and Sumy = V1y + V2y

V1 is the velocity of the airplane and V2 is the velocity of the wind.

- - 92 degrees
- 177 degrees
- -52 degrees
- 88 degrees
- - 88 degrees

Question 20 4 pts

hide the hint.

An object has a position given by $\vec{r} = [2.0\text{ m} + (5.00\text{ m/s})t]\hat{i} + [3.0\text{ m} - (2.00\text{ m/s}^2)t^2]\hat{j}$, where quantities are in SI units. What is the speed of the object at time $t = 2.00$ s?

hide the hint.

hint: first take the derivative to find the component of the instantaneous velocity as a function of time

V = Vx(t) i + Vy(t) j. Then plug time to get the components of V(t=2s). Then find the magnitude of this vector. This is the speed.

- 9.43m/s
- 7.0 m/s
- 6.40m/s
- 13m/s

Question 21 4 pts

Object A has a position as a function of time given by $\vec{r}_A(t) = (3.00\text{ m/s})t\,\hat{i} + (1.00\text{ m/s}^2)t^2\hat{j}$. Object B has a position as a function of time given by $\vec{r}_B(t) = (4.00\text{ m/s})t\hat{i} + (-1.00\text{ m/s}^2)t^2\hat{j}$. All quantities are SI units. What is the distance between object A and object B at time $t = 3.00$ s?

hide the hint first

hint: first find the components of points A and B at t=3s. (plug t=3s). Then find the components of the displacement C = B - A . C (xB-xA, yB-yA). The magnitude of this vector is the distance (pythagorean ..)

- 15m
- 3.46m
- 18.3m
- 29.8m

○ 34.6m

Question 22 5 pts

An object has a position given by $\vec{r} = [2.0\ \text{m} + (3.00\ \text{m/s})t]\hat{i} + [3.0\ \text{m} - (2.00\ \text{m/s}^2)t^2]\hat{j}$, where all quantities are in SI units. What is the magnitude of the acceleration of the object at time $t = 2.00$ s?

hide the hint

First find the components of the acceleration as a function of time. a=ax(t) i + ay(t) j . To get ax(t) derive twice x(t) and to get ay(t) derive twice y(t). Then plug t=2s to get the components of a(t=2s). Find the magnitude of this vector.

○ 0.522m/s/s

○ 2m/s/s

○ 4m/s/s

○ 0m/s/s

○ 1m/s/s

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